



Artificial intelligence-enabled electrocardiograms for identification of patients with low ejection fraction: a pragmatic, randomized clinical trial

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We have conducted a pragmatic clinical trial aimed to assess whether an electrocardiogram (ECG)-based, artificial intelligence (AI)-powered clinical decision support tool enables early diagnosis of low ejection fraction (EF), a condition that is underdiagnosed but treatable. In this trial (NCT04000087), 120 primary care teams from 45 clinics or hospitals were cluster-randomized to either the intervention arm (access to AI results; 181 clinicians) or the control arm (usual care; 177 clinicians). ECGs were obtained as part of routine care from a total of 22,641 adults (N = 11,573 intervention; N = 11,068 control) without prior heart failure. The primary outcome was a new diagnosis of low EF ($\leq 50\%$) within 90 days of the ECG. The trial met the prespecified primary endpoint, demonstrating that the intervention increased the diagnosis of low EF in the overall cohort (1.6% in the control arm versus 2.1% in the intervention arm, odds ratio (OR) 1.32 (1.01–1.61), $P = 0.007$) and among those who were identified as having a high likelihood of low EF (that is, positive AI-ECG, 6% of the overall cohort) (14.5% in the control arm versus 19.5% in the intervention arm, OR 1.43 (1.08–1.91), $P = 0.01$). In the overall cohort, echocardiogram utilization was similar between the two arms (18.2% control versus 19.2% intervention, $P = 0.17$); for patients with positive AI-ECGs, more echocardiograms were obtained in the intervention compared to the control arm (38.1% control versus 49.6% intervention, $P < 0.001$). These results indicate that use of an AI algorithm based on ECGs can enable the early diagnosis of low EF in patients in the setting of routine primary care.

A symptomatic left ventricular systolic dysfunction (ALVSD) is present in 1.4–2.2% of the general population and confers a 6.5-fold increased risk of clinical heart failure (HF) and a 1.6-fold increased risk of all-cause mortality^{1–5}. Although guideline-recommended therapies can effectively reduce the risk of HF progression and mortality, low ejection fraction (EF) is often undiagnosed, leading to missed opportunities for treatment^{6–8}.

EF can be readily determined by echocardiography, but the imaging test is too resource-intensive to screen unselected populations. A low-cost, widely available test could be used to identify high-risk patients for further assessment with echocardiography⁹. However, no such test has demonstrated sufficient performance to make this approach feasible^{10–16}.

An artificial intelligence (AI) algorithm was recently developed to identify patients at a high likelihood of low EF based on a standard 12-lead electrocardiogram (ECG) in a sample of 44,959 patients with paired ECGs and echocardiograms¹⁷. The algorithm uses neural networks that recognize subtle patterns in complex

signals to extract meaningful information. In a prospective validation, the algorithm demonstrated excellent performance with a C-statistic of 0.92¹⁸. However, it is unknown whether the AI, when integrated into routine clinical workflows, can improve practice. The ECG AI-Guided Screening for Low Ejection Fraction (EAGLE) trial thus aimed to assess whether the AI enables the early diagnosis of low EF in everyday practice.

Clinician and patient characteristics. The characteristics of clinicians were similar in the intervention and control arms. The mean age of clinicians was 45.5 years (standard deviation (s.d.) 10.8), 58.6% were women, 91.7% were White, 65.5% were physicians, 12.4% were physician assistants and 22.1% were nurse practitioners (Supplementary Table 1).

A total of 22,641 adults (11,068 in the control group and 11,573 in the intervention group) were included in the analyses (Fig. 1). The mean age was 60.5 ± 17.6 years and 53.9% were women (Table 1). The most common indications for ECG were ‘chest pain’ and ‘baseline screening’ (Supplementary Table 2).

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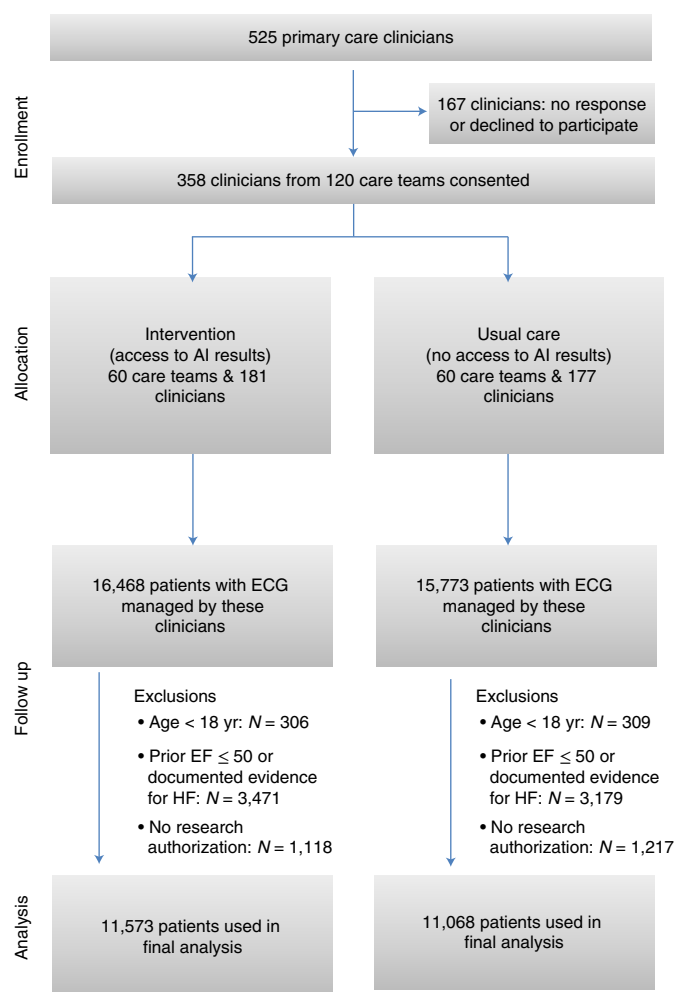


Fig. 1 | CONSORT flow diagram.

The AI-ECG was positive in 664 patients (6.0%) in the control group and 692 (6.0%) in the intervention group. Older adults, men, non-White patients, patients with hypertension, diabetes, MI, AF, valvular heart disease, CKD, prior echocardiogram and those whose ECGs were ordered in the emergency room or hospital (rather than the outpatient clinic) were more likely to receive a positive result (Supplementary Table 3).

Echocardiogram performed after ECGs. In the overall population, the proportion of patients who underwent echocardiography was not different between groups (18.2% in the control group versus 19.2% in the intervention group, OR 1.08 (0.97–1.20), $P=0.17$). The median time between ECG and echocardiogram was eight days (interquartile range (IQR) 1–29). Older adults, patients with comorbidities (hypertension, MI, AF and valvular heart disease) and those whose ECGs were ordered in the hospital were more likely to receive echocardiograms, whereas women and those living in rural areas were less likely to receive them (Supplementary Table 4).

Among those who had a positive AI result, a greater proportion of patients in the intervention group underwent echocardiography (38.1% in the control group versus 49.6% in the intervention group, OR 1.63 (1.24–2.13), $P<0.001$). In patients in the intervention group who had a positive AI-ECG result, those aged 65–74 years and those with MI or AF were more likely to undergo echocardiography, whereas those who had prior echocardiograms were less likely to undergo echocardiography (Supplementary Table 5).

Clinical outcomes. In the overall cohort, the intervention increased the diagnosis of low EF from 1.6% in the control group to 2.1% in the intervention group (OR, 1.32 (1.08–1.61), $P=0.007$). Among 1,356 patients who had a positive result, the intervention improved the diagnosis of low EF from 14.5% in the control group to 19.5% in the intervention group (OR 1.43 (1.08–1.91), $P=0.01$). The observed intracluster correlation coefficient in this trial was 0.0017.

Results using different thresholds for low EF were consistent with the primary result. For example, in the overall cohort, the diagnosis of EF $\leq 40\%$ was numerically higher in the intervention group but not statistically significant (0.6% in the control group versus 0.9% in the intervention group; OR 1.39 (0.99, 1.95), $P=0.057$; Supplementary Table 6).

Among 172 patients with a newly diagnosed EF $\leq 40\%$, 38.4% were already treated with angiotensin-converting-enzyme inhibitors (ACEi) or angiotensin II receptor blocker (ARB), and 31.4% were treated with beta blockers at baseline. During the 90-day follow-up, 73.3% received at least one new prescription of ACEi, ARB or beta blocker. Combining baseline and follow-up prescriptions, 95.3% of patients received at least one prescription, with 79.1% receiving ACEi or ARB, and 91.3% receiving beta blockers (Supplementary Table 7).

Subgroup and secondary analyses. The impact of the intervention on newly diagnosed low EF was largely consistent across subgroups. There was only one significant interaction: the intervention increased the diagnosis of low EF to a greater degree in those whose ECGs were performed in outpatient clinics (Fig. 2). For patients who had an ECG performed in the outpatient setting, the intervention increased the diagnosis of low EF (1.0% for control versus 1.6% for intervention, OR 1.71 (1.23–2.37), $P=0.001$).

In the overall cohort, the impact on the use of echocardiography was consistent across the subgroups (Extended Data Fig. 1). Among patients with positive results, the intervention increased the use of echocardiography more significantly in women versus men (P for interaction, 0.02), in patients without prior echocardiogram versus those with prior echocardiogram (P for interaction, 0.04) and in patients whose ECGs were ordered in outpatient clinics (P for interaction, 0.02; Extended Data Fig. 2).

Among the 4,239 patients who underwent an echocardiogram, 419 (9.9%) had other clinically important findings, most of which (390 (9.2%)) were moderate or severe valvular heart disease (Supplementary Table 8). The intervention did not change the diagnosis of other findings in the overall population (Supplementary Table 9), but increased the diagnosis in patients with a positive AI-ECG result (5.9% for control versus 9.4% for intervention, OR 1.63 (1.04–2.56), $P=0.03$; Supplementary Table 10).

Discussion

Although many AI algorithms for medical applications have been developed and validated, very few have been prospectively evaluated in randomized controlled trials (RCTs). The current study is one of the first RCTs to assess an AI-powered clinical decision support tool in routine practice. The trial demonstrates an increased identification of low EF, a condition that is often asymptomatic and underdiagnosed in its early stage. By creating an information technology infrastructure to automatically analyze ECGs and report AI results to clinicians in real time, the new tool represents an opportunity to identify and treat low EF earlier and thus potentially reduce the disease burden and mortality in broad populations. However, we emphasize that further work is needed to establish the cost-effectiveness of this approach and to estimate the downstream clinical impact of these new diagnoses.

The trial was conducted across diverse practice settings, including primary care clinics at a tertiary medical center as well as community and rural clinics. The clinicians involved in the

Table 1 | Baseline characteristics of the patients

Characteristic	Control (N = 11,068)	Treatment (N = 11,573)	Overall (N = 22,641)
Age (yr, mean (s.d.))	60.5 (17.6)	60.5 (17.5)	60.5 (17.6)
18–64 yr, n (%)	5,934 (53.6)	6,256 (54.1)	12,190 (53.8)
65–74 yr, n (%)	2,630 (23.8)	2,764 (23.9)	5,394 (23.8)
≥75 years, n (%)	2,504 (22.6)	2,553 (22.1)	5,057 (22.3)
Female, n (%)	6,123 (55.3)	6,080 (52.5)	12,203 (53.9)
Race, n (%)			
White	10,423 (94.2)	10,926 (94.4)	21,349 (94.3)
Black or African American	214 (1.9)	201 (1.7)	415 (1.8)
Asian	136 (1.2)	145 (1.3)	281 (1.2)
Other	237 (2.1)	254 (2.2)	491 (2.2)
Unknown/declined/missing	58 (0.5)	47 (0.4)	105 (0.5)
Rural, n (%)	5,019 (45.4)	6,323 (54.6)	11,342 (50.1)
Medical history, n (%)			
Hypertension	6,177 (55.8)	6,491 (56.1)	12,668 (56.0)
Diabetes	2,221 (20.1)	2,347 (20.3)	4,568 (20.2)
Myocardial infarction	717 (6.5)	770 (6.7)	1,487 (6.6)
Peripheral artery disease	444 (4.0)	411 (3.6)	855 (3.8)
Stroke or TIA	381 (3.4)	409 (3.5)	790 (3.5)
Prior AF	919 (8.3)	991 (8.6)	1,910 (8.4)
New AF on index ECG	248 (2.2)	246 (2.1)	494 (2.2)
Valvular heart disease	152 (1.4)	129 (1.1)	281 (1.2)
Chronic kidney disease	1,209 (10.9)	1,373 (11.9)	2,582 (11.4)
Prior echocardiogram	1,896 (17.1)	1,903 (16.4)	3,799 (16.8)
Location of ECG ordered, n (%)			
Outpatient clinic	5,969 (53.9)	6,043 (52.2)	12,012 (53.1)
Emergency room	4,056 (36.6)	4,411 (38.1)	8,467 (37.4)
Hospital	1,043 (9.4)	1,119 (9.7)	2,162 (9.5)

AF, atrial fibrillation; TIA, transient ischemic attack

study were not AI experts or specialists but were busy clinicians engaged in the care of patients presenting with a wide range of conditions. Therefore, this trial might form a template for how new technologies can be studied in ways that closely mirror their potential deployment.

The addition of AI to ECG stands to add value to an existing inexpensive and widely used test, but the impact depends on several factors. First, the tool might be more useful in populations with higher rates of missed or delayed diagnoses. We demonstrate that the diagnostic yield was more pronounced for patients whose ECGs were obtained in outpatient clinics than hospitalized patients. This may be because low EF is relatively underdiagnosed in outpatient settings in comparison to hospital settings where the use of echocardiography is more frequent. Similarly, in a resource-constrained environment, such a tool may help clinicians allocate scarce resources more effectively.

The impact is also dependent on whether clinicians respond to the AI recommendations. In the current study, the intervention increased the use of echocardiograms from 38.1% to 49.6%. The relatively modest increase might be because the intervention was deployed in a health system that adopts value-based health delivery models and emphasizes judicious use of imaging services, such as echocardiography. A more prescriptive recommendation or even a cascading order set may have increased echocardiography utilization and the diagnostic yield of low EF even further.

The impact of the tool also depends on the cut-point selected to trigger clinician action. In the current study, we selected the cut-point used in the initial derivation study that optimized the sensitivity and specificity equally. Only 6% of the patients had a positive AI-ECG result, and the overall use of echocardiography was not different between intervention and control groups, providing some reassurance that even this relatively sensitive threshold would not increase overall healthcare utilization. A more specific cut-point could reduce healthcare utilization and false positive rates, but at the expense of lower sensitivity. In addition, the model could be used to identify patients who may be candidates for particular therapies ($EF \leq 40\%$ to define HFrEF or $EF \leq 35\%$ to identify patients who could benefit from implantable cardioverter defibrillator (ICD) implantation, for instance). Another alternate is to present the predicted risk as a continuous variable and allow clinicians to make their own decisions regarding the appropriate threshold for ordering an echocardiogram on a case-by-case basis. However, in a feasibility study conducted before the trial, most clinicians interviewed had indicated that, because there are no well-established norms for how to interpret a continuous AI risk/probability, a binary recommendation statement was preferred.

This pragmatic trial also establishes a framework for rapid evaluation and implementation of AI algorithms within existing electronic health records (EHRs). Traditional RCTs are resource-intensive and time-consuming, and are conducted in

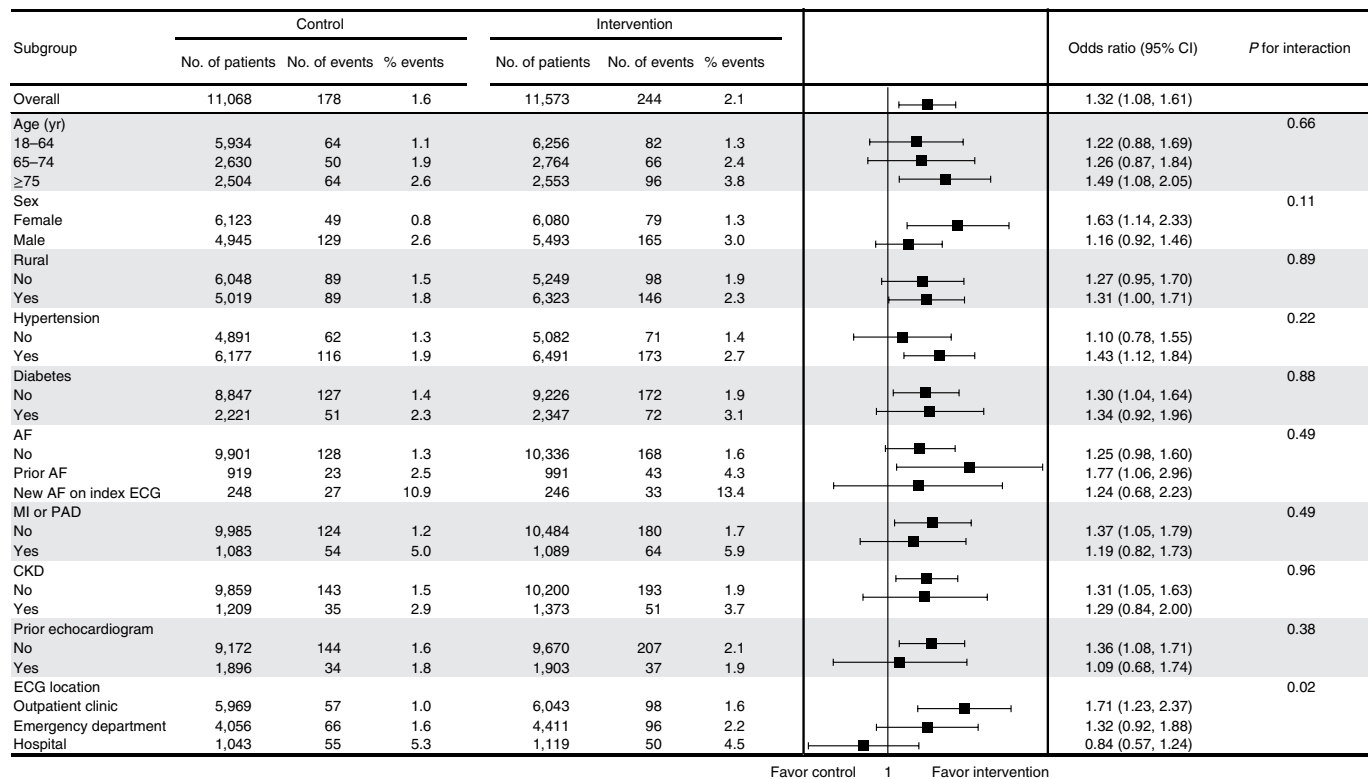


Fig. 2 | Subgroup analyses for primary endpoint. Error bars indicate 95% confidence intervals. Mixed-effect logistic regressions were used for the statistical test; this was two-sided, with no adjustment for multiple comparison.

highly controlled environments in selected patient populations. By contrast, this pragmatic trial was embedded in everyday practice to efficiently generate relevant and valid evidence. The study design, embedding of the technology in the EHR, and enrollment and follow-up were completed in less than two years at a much lower cost than a traditional RCT. Given the increasing number of AI tools being developed, this framework might facilitate rigorous and timely evaluation and implementation.

The conduct of this trial was feasible because of the well-curated infrastructure that enables access to EHR data longitudinally and in real time for analytics¹⁹. A variety of digital phenotyping algorithms that serve as clinical decision support and quality measurement tools already existed in our practice and can be used to determine a patient's medical history and clinical outcomes. Evidence generated from the trial was reported to practice leadership to facilitate the enterprise-wide implementation of the AI algorithm. Therefore, the current work represents an example of a learning health system that forms a feedback cycle for continuous learning and improvement²⁰.

The study has several limitations. First, in this trial, clinicians received alerts and reminders as well as encouragement from practice leadership and local champions. When implemented in other practices, if no such support is provided, the improvement might be smaller. Second, nearly all the patients had insurance coverage and few clinicians suggested that the cost of the echocardiogram was the reason for not ordering an echocardiogram. However, in uninsured or under-insured populations, out-of-pocket costs could prevent patients from undergoing an echocardiogram. Third, the study was not designed to determine the long-term clinical impact, for example, HF hospitalizations and mortality. However, the reductions in morbidity and mortality with pharmacologic and non-pharmacologic therapy in patients with ALVSD have already been demonstrated in RCTs and are recommended in guidelines^{6–8}.

An AI algorithm run on existing ECGs enabled the early diagnosis of low EF in a large cohort of patients managed in routine primary care practices. Because ECG is a low-cost test that is frequently performed for a variety of purposes, the algorithm could potentially improve early diagnosis and treatment of a condition that is often asymptomatic but has effective treatments, and thus reduce the disease burden in broad populations.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-021-01335-4>.

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Methods

Trial design. We conducted a pragmatic, cluster RCT in 45 Mayo Clinic primary care practices, including an academic medical center as well as community and rural clinics across Minnesota and Wisconsin.

The study participants were primary care clinicians, including physicians, nurse practitioners and physician assistants, organized as teams that care for adult patients (≥ 18 years). Clinicians in the pediatric care teams, resident care teams, acute and urgent care teams, and nursing home care teams were excluded.

The trial was approved by the Institutional Review Board at Mayo Clinic. The first invitation was sent to eligible clinicians on 29 June 2019 and the enrollment period ended on 26 July 2019. A total of 358 of 525 eligible primary care clinicians provided informed consent. No clinicians withdrew consent during the study period. Patients themselves were not considered 'subjects' in this study (as such, patients themselves were not consented for participation). Patients' data were collected using EHRs. A patient's data were included in the analysis if the patient received an ECG for any indication between 5 August 2019 and 31 March 2020. Patients were excluded if they were under 18 years old, had known EF $\leq 50\%$ or a history of documented HF identified by clinical notes abstracted by natural language processing (NLP)²¹ or they did not provide authorization to use their data for research.

The trial was registered before initiation (ClinicalTrials.gov number NCT04000087), and the study protocol and materials have been published previously^{22,23}. We followed the guidelines outlined in the CONSORT-AI Extension checklist²⁴ in the reporting of the study.

Randomization. Randomization was conducted at the care team level, with each care team serving as a cluster. Primary care at Mayo Clinic is delivered by care teams, with physicians, advanced practice providers (nurse practitioners and physician assistants), nurses and other healthcare professionals organized into discrete functional units. The randomization was stratified by region (Rochester, Southeast Minnesota, Southwest Minnesota, Northwest Wisconsin and Southwest Wisconsin) and patient volume (that is, the number of potential trial-eligible patients within the first six months of 2019). A total of 181 clinicians from 60 care teams were randomized to the intervention group, and 177 clinicians from 60 care teams were randomized to the control/usual care group.

Treatment. We employed the same algorithm as was reported previously¹⁷. The algorithm was run on the digital ECG files housed in our institutional data lake to generate a report every time an ECG was performed. The AI was run on ECGs that were obtained and reported for routine clinical practice. Only interpretable studies were processed by the centralized ECG laboratory. Each ECG was interpreted first by a CRAT-certified technologist (CRAT, Certified Rhythm Analysis Technician) and then over-read by a cardiologist. Uninterpretable or poor-quality ECGs were not entered into the EHR. As such, the AI was run on all ECGs that were felt to be of adequate quality for routine clinical use.

Primary care clinicians in the intervention group had access to the report, which displayed whether the AI-ECG result was positive or negative. When the AI-ECG result was negative, the recommendation was 'no further testing unless indicated by other symptoms or conditions'. When the AI-ECG result was positive, the recommendation was to 'consider ordering an echocardiogram' (Extended Data Fig. 3). The report was designed using an iterative, user-centered approach, during which a series of prototypes were developed, tested and revised based on interviews with 16 clinicians from diverse disciplines. An internal website was built to provide videos of trial investigators explaining the AI-ECG and the study procedures, frequently asked questions, publications and a patient brochure that can be used to explain AI-ECG results to patients (Extended Data Fig. 4).

During the trial, clinicians in the intervention group received an alert when their patients had a positive result. If they opted not to order an echocardiogram, they were asked to provide an explanation via a hyperlinked multiple-choice questionnaire. If no echocardiogram was ordered and no reason was provided, clinicians would receive a weekly email reminder. Clinicians also received a monthly report summarizing the AI-ECG results at the care team level, including the proportion of patients with a positive AI-ECG result, receiving an echocardiogram, diagnosed with low EF and receiving guideline-recommended therapies. Clinicians in the usual care group did not receive any contact from the investigators.

Baseline characteristics. A patient's ECG date was defined as the index date. All the EHR before this date were used to determine a patient's medical history. Hypertension, diabetes, myocardial infarction (MI), peripheral artery disease (PAD), stroke or transient ischemic attack (TIA), and atrial fibrillation (AF) were determined using validated NLP programs to abstract clinical notes. Valvular heart disease and chronic kidney disease (CKD) were determined using diagnosis codes. We also examined prescription records over the past 12 months to determine whether patients were treated with ACEi, ARBs and beta blockers.

Outcomes. The primary outcome was newly diagnosed low EF, defined as EF $\leq 50\%$ within 90 days. Secondary outcomes included the completion of an echocardiogram within 90 days and other findings (for example, valvular heart

disease), except low EF, present on echocardiograms. Sensitivity analyses were conducted using different thresholds to define low EF, for example $<50\%$, $\leq 40\%$ (the cut-point that defines heart failure with reduced ejection (HFrEF) fraction in current guidelines) and $\leq 35\%$ (the cut-point that is often used for primary prevention implantable cardioverter defibrillator (ICD) implantation). Among patients with newly discovered EF $\leq 40\%$, we examined prescriptions of ACEi, ARBs and beta blockers. These are considered as exploratory outcomes. Secondary outcomes included in the protocol but were not reported here included long-term health utilization, cost-effectiveness and implementation strategies, which will be published subsequently as separate studies as they require more time for data collection and analysis.

Sample size. Our preliminary analysis using 1,234 primary care patients who underwent an ECG in September 2018 found 2.4% had a primary endpoint. Given that we expected to have about 60 care teams per arm, we set our target as 12,000 patients per arm (that is, an average of 200 ECGs per care team). If the intraclass correlation coefficient is 0.01, we would have 80% power to detect a significant difference if low EF is diagnosed in 3.45% of the intervention group (a relative increase of 44%) assuming a two-tailed test with a type I error of 0.05. If the intraclass correlation coefficient is 0.03, we will have 80% power if low EF is diagnosed in 4.06% of the intervention group (a relative increase of 69%). The trial started on the first Monday of August when the clinical decision support was turned on for clinicians in the intervention group and ended on 31 March 2020 when the number of patients per arm was quite close to 12,000 patients per arm.

Statistical analysis. Mixed-effect logistic regression was used to compare the intervention group to the control group with the care team as a random effect. To perform the mixed-effect logistic regression, the R package lme4, specifically the glmer function, can be used; the equivalent Stata package is melogit. Subgroup analyses were performed by age, sex, baseline comorbidities, whether or not patients had previously received an echocardiogram, and where the ECG was ordered (outpatient clinic, emergency department or hospital). The subgroup effects, that is, whether the ORs were the same across subgroups, were tested by the significance of the interaction term added to the regression model. A two-sided *P* value of less than 0.05 was considered significant. We also used mixed-effect multivariable logistic regression to assess characteristics associated with getting a positive AI result or getting an echocardiogram. All the analyses except those related to the primary outcome were considered exploratory. Data management was conducted using SAS 9.4 software (SAS Institute) and analyses were performed using Stata 14.1 (Stata Corp) and RStudio 1.2.5033.1 (RStudio Team, 2019).

Role of funding source. The Mayo Clinic Robert D. and Patricia E. Kern Center for the Science of Health Care Delivery provided funding for the study, but played no role in the study design, the collection, analysis and interpretation of data, in the writing of the report, or in the decision to submit the paper for publication. All authors had full access to the data in the study and accept responsibility to submit for publication.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this Article.

Data availability

The data are not publicly available because they are electronic health records, consented for research use by Mayo Clinic investigators. Making the data publicly available without additional consent or ethical approval might compromise patients' privacy and the original ethical approval. If other investigators are interested in performing additional analyses, requests can be made to the corresponding author (X.Y.) and analyses will be performed in collaboration with the Mayo Clinic.

Code availability

The AI algorithm, which was previously published, cannot be made publicly available because it is proprietary intellectual property (patent pending). The AI algorithm cannot be used in routine practice before getting FDA approval, and this algorithm is currently undergoing a submission/review process with the FDA. The AI algorithm is available upon request for research studies.

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Competing interests

Mayo Clinic has licensed the AI-ECG algorithm to EKO, a maker of digital stethoscopes with embedded ECG electrodes. At no point will Mayo Clinic benefit financially from the use of the AI-ECG for the care of patients at the Mayo Clinic. P.A.F., F.L.-J., S.K. and Z.I.A. may receive financial benefit from this agreement for use of the AI-ECG outside of the Mayo Clinic. All other authors declare no competing interests.

Additional information

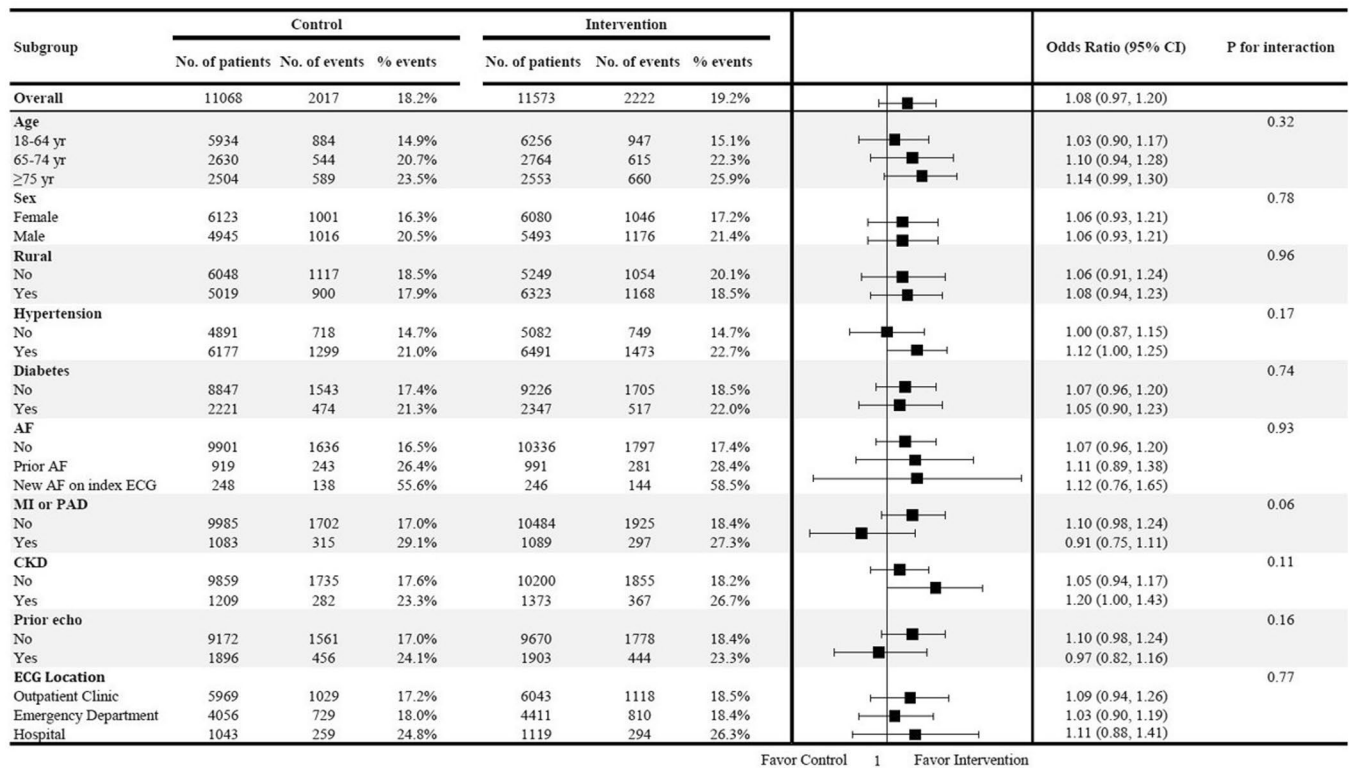
Extended data is available for this paper at <https://doi.org/10.1038/s41591-021-01335-4>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41591-021-01335-4>.

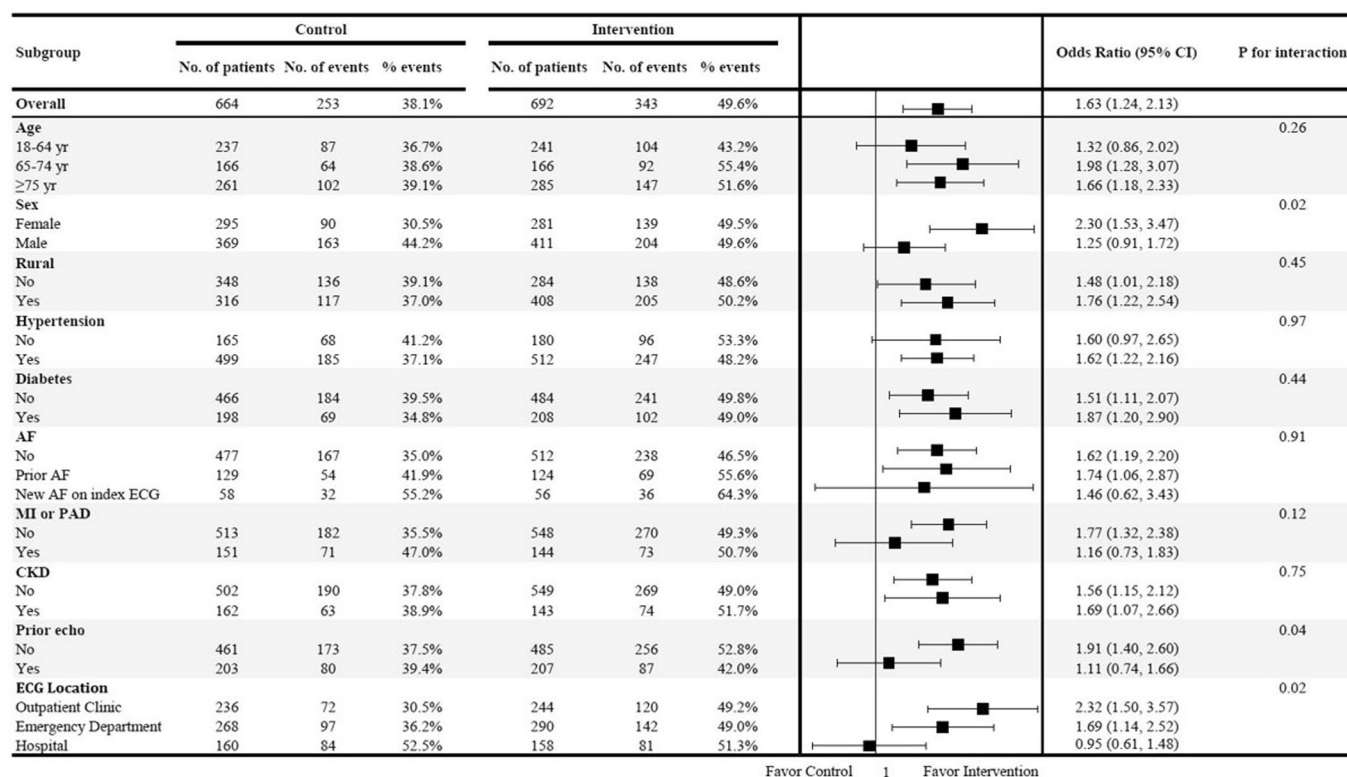
Correspondence and requests for materials should be addressed to X.Y.

Peer review information *Nature Medicine* thanks Jill Waalen, Giorgio Quer and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Michael Basson was the primary editor on this article and managed its editorial process and peer review in collaboration with the rest of the editorial team.

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Extended Data Fig. 1 | Subgroup analyses for echocardiogram performed. Outcome was echocardiogram being performed. An odds ratio greater than 1 means a higher likelihood of echocardiogram being performed. Error bars indicate 95% confidence intervals; mixed effect logistic regressions were used for the statistical test; it was two-sided; no adjustment for multiple comparison was made.



Extended Data Fig. 2 | Subgroup analyses for echocardiogram performed among patients with positive screening results. Outcome was echocardiogram being performed. An odds ratio greater than 1 means a higher likelihood of echocardiogram being performed. Error bars indicate 95% confidence intervals; mixed effect logistic regressions were used for the statistical test; it was two-sided; no adjustment for multiple comparison was made.

Test Patient Age: XX Sex: X DOB: X MC#: XXXXXX					
ECG AI Guided Screening for Low Ejection Fraction (EAGLE)					
ALGORITHMIC RESULTS					
Screening result: NEGATIVE Recommendation: No further testing unless indicated by other symptoms or conditions *Results generated from ECG-based AI algorithm AskMayoExpert: Reduced Ejection Fraction ECG Lab: 555-555-555	The algorithm is being applied in order to screen for asymptomatic left ventricular systolic dysfunction in patients who have no other indication for echocardiography. Among patients with established heart failure, heart failure symptoms, or other indications for an echocardiogram, this algorithm should not affect your decision to order an echocardiogram. Risk factors for heart failure, such as hypertension, obesity, diabetes, dyslipidemia, atherosclerotic disease, smoking, and alcohol abuse, should also be evaluated and managed to prevent heart failure. The prediction algorithm was derived from a sample of Mayo Clinic patients who underwent both ECG and echocardiography. The model demonstrated a c statistic of 0.92, a sensitivity of 82.5%, a specificity of 86.8%, and an accuracy of 86.5% in a prospective validation.				
Test Patient Age: XX Sex: X DOB: X MC#: XXXXXX					
ECG AI Guided Screening for Low Ejection Fraction (EAGLE)					
ALGORITHMIC RESULTS					
Screening result: POSITIVE Recommendation: Consider ordering an echocardiogram *Results generated from ECG-based AI algorithm AskMayoExpert: Reduced Ejection Fraction ECG Lab: 555-555-555	The algorithm is being applied in order to screen for asymptomatic left ventricular systolic dysfunction in patients who have no other indication for echocardiography. Among patients with established heart failure, heart failure symptoms, or other indications for an echocardiogram, this algorithm should not affect your decision to order an echocardiogram. Risk factors for heart failure, such as hypertension, obesity, diabetes, dyslipidemia, atherosclerotic disease, smoking, and alcohol abuse, should also be evaluated and managed to prevent heart failure. The prediction algorithm was derived from a sample of Mayo Clinic patients who underwent both ECG and echocardiography. The model demonstrated a c statistic of 0.92, a sensitivity of 82.5%, a specificity of 86.8%, and an accuracy of 86.5% in a prospective validation.				
RECENT CV HISTORY					
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Prev Echo	01/01/03				
Stress Echo	01/01/03				
SCHEDULED CV PROCEDURES & TESTS					
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Echo	05/05/19				

Extended Data Fig. 3 | Clinician-facing action recommendation report for screening results.

AI ECG

[Home](#) » [Innovation](#) » AI ECG

Artificial-Intelligence Enhanced Electrocardiogram

The application of AI to ECG may help facilitate the current ECG interpretation workflow, allow the ECG lab to scale its clinical volume, and improve the accuracy and consistency of ECG interpretation. Furthermore, AI may allow us to identify patterns not readily apparent to a human reader (e.g. identification of low ejection fraction). We anticipate these technologies may allow us to reach new patients through relationships with new Mayo Clinic clinical partners and consumer-facing applications.

- [ECG Lab As Platform for Innovation](#)  - Dr. Peter Noseworthy at the Quarterly Administrative Update
- [AI-Enhanced ECG Interpretation: Research to Practice Innovation](#)  - Dr. Peter Noseworthy at the Research Gathering



EAGLE

BEAGLE

ECG AI-Guided Screening for Low Ejection Fraction (EAGLE)

A deep learning algorithm, built in Epic, automatically screens for low ejection fraction and encourages clinicians to obtain a confirmatory transthoracic echocardiogram for previously undiagnosed patients, thereby facilitating early identification and prevention of heart failure. A large RCT is underway in all primary care practices in the Mayo Clinic Health System and Rochester.

[FAQ](#)[Where to Find AI-Enhanced ECG Report](#)

Videos

- [Dr. Xiaoxi Yao Describes the EAGLE Trial Procedures](#) 
- [Dr. Peter Noseworthy Describes the EAGLE Algorithm](#) 
- [Dr. Rozalina McCoy Explains How to Find AI-Enhanced ECG Report](#) 
- [Dr. Rozalina McCoy Discusses the Value of the EAGLE Algorithm from a Primary Care Provider Perspective](#) 
- [CNBC Interview with Dr. Paul Friedman about EAGLE Algorithm](#)

Resources

[Patient Brochure](#) [Development of EAGLE Algorithm
Published in Nature Medicine](#) [Validation of EAGLE Algorithm
Published in Journal of Cardiovascular
Electrophysiology](#) 

Extended Data Fig. 4 | Internal resources website for clinicians.

Reporting Summary

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- | | | |
|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☒ | ☐ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ | The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☐ | ☒ | A description of all covariates tested |
| ☐ | ☒ | A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection Computer code is available in the Appendix.

Data analysis Data management was conducted using SAS 9.4 (SAS Institute Inc., Cary, North Carolina) and analyses were performed using Stata 14.1 (Stata Corp, College Station, TX) and RStudio 1.2.5033.1 (RStudio Team [2019], including the R package lme4).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Data availability statements

The dataset is not publicly available since they are electronic health records, consented for research use by Mayo Clinic investigators. Making the data publicly available without additional consent or ethical approval might compromise patients' privacy and the original ethical approval. If other investigators are interested in performing additional analyses, requests can be made to the corresponding author X.Y. and analyses will be performed in collaboration with Mayo Clinic. An alternative is to contact patients to obtain additional consent.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The sample size is efficient based on this power calculation: our preliminary analysis using 1,234 primary care patients who underwent an ECG in September 2018 found in 2.4% had a primary endpoint. Since we expected to have about 60 care teams per arm, we set our target as 12,000 patients per arm (i.e., an average of 200 ECGs per care team). If the intracluster correlation coefficient is 0.01, we would have 80% power to detect a significant difference if low EF is diagnosed in 3.45% of the intervention group (a relative increase of 44%) assuming a 2-tailed test with a type I error of 0.05. If the intracluster correlation coefficient is 0.03, we will have 80% power if low EF is diagnosed in 4.06% of the intervention group (a relative increase of 69%). The trial started on the first Monday of August when the clinical decision support was turned on for clinicians in the intervention group and ended on 3/31/2020 when the number of patients per arm was quite close to 12,000 patients per arm.
Data exclusions	A patient's data were included in the analysis if the patient received an ECG for any indication between August 5th, 2019 and March 31st, 2020. Patients were excluded if they were under 18 years old, had known EF≤50% or a history of documented HF or they did not provide authorization to use their data for research. The exclusion criteria were pre-established.
Replication	There were no replication attempts for this prospective randomized trial due to cost and time requirements.
Randomization	Randomization was conducted at the care team level, with each care team serving as a cluster. Primary care at Mayo Clinic is delivered by care-teams, with physicians, advanced practice providers (APPs; nurse practitioners and physician assistants), nurses, and other health care professionals organized into discrete functional units. The randomization was stratified by region (Rochester, Southeast Minnesota, Southwest Minnesota, Northwest Wisconsin, and Southwest Wisconsin), and patient volume (i.e., the number of potential trial-eligible patients within the first six months of 2019). A total of 181 clinicians from 60 care teams were randomized to the intervention group, and 177 clinicians from 60 care teams were randomized to the control/usual care group.
Blinding	The study was not blinded due to the nature of the intervention. Clinicians either had access to artificial intelligence results of patients or not, which could not be blinded.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems	Methods
n/a	Involvement in the study
☒ ☐ Antibodies	☒ ☐ ChIP-seq
☒ ☐ Eukaryotic cell lines	☒ ☐ Flow cytometry
☒ ☐ Palaeontology and archaeology	☒ ☐ MRI-based neuroimaging
☒ ☐ Animals and other organisms	
☐ ☒ Human research participants	
☐ ☒ Clinical data	
☒ ☐ Dual use research of concern	

Human research participants

Policy information about [studies involving human research participants](#)

Population characteristics	The mean age of clinicians was 45.5 years (standard deviation [SD] 10.8), 58.6% were women, 91.7% were white, 65.5% were physicians, 12.4% were physician assistants, and 22.1% were nurse practitioners. A total of 22,641 adult patients (11,068 patients in the control group and 11,573 patients in the intervention group) were included in the analyses. The mean age was 60.5±17.6 years; 53.9% were women.
Recruitment	We sent clinicians at eligible care teams a recruitment email that included a description of the intervention and the trial and asked them to reply to indicate consent for participation. The email explained that participation is voluntary and choosing to participate or not would not affect any aspect of their employment or work performance evaluation. Clinician champions may also help raise awareness of the trial at their site(s) through e-mail, presentations at department meetings, and word of

mouth. We sent another three follow-up email reminders (not including those e-mails from local champions) to clinicians who did not respond. Clinicians who were not interested in artificial intelligence or who were busier might tend to decline participation, and if so, the true effect might be smaller than what was observed

Ethics oversight

The trial was approved by the institutional review board at Mayo Clinic.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration ClinicalTrials.gov Identifier: NCT04000087

Study protocol

Yao, X., et al. ECG AI-Guided Screening for Low Ejection Fraction (EAGLE): Rationale and design of a pragmatic cluster randomized trial. American heart journal 219, 31-36 (2019).

Yao, X., et al. Clinical trial design data for electrocardiogram artificial intelligence-guided screening for low ejection fraction (EAGLE). Data in brief, 104894 (2019).

Data collection

The trial was conducted across diverse practice settings, including primary care clinics at a tertiary medical center as well as community and rural clinics. Patients' data were collected from the electronic health records and clinicians' data were self reported via electronic questionnaires. Clinicians were consented between 6/26/2019-7/30/2019. A patient's data were included in the analysis if the patient received an ECG for any indication between August 5th, 2019 and March 31st, 2020. Outcome data were collected within 90 days of the ECG.

Outcomes

The primary outcome was newly diagnosed low EF, defined as $EF \leq 50\%$ within 90 days. Sensitivity analyses were conducted using different thresholds to define low EF, e.g. $\leq 40\%$ and $\leq 35\%$. Secondary outcomes included the completion of an echocardiogram within 90 days and other findings (e.g., valvular heart disease), except low EF, present on echocardiograms. Among patients with newly discovered $EF \leq 40\%$, we examined prescriptions of ACEi, ARBs, and beta blockers.